

SUMMARY

Software Development Engineer (Full Stack) with over 4 years of experience across web and mobile product development. Proficient in MEAN/MERN, Flutter/NestJS, and Python/Flask stacks, with hands-on GIS and geospatial data experience. Skilled in end-to-end development, debugging, testing, and cloud deployments across collaborative and freelance projects.

EXPERIENCE

Phicode Pvt. Ltd. - Software Development Engineer Aug 2022 - Present

Bengaluru, Karnataka

Telecom Tower Management Application

- Built a telecom tower management platform (Angular, Express.js) that lets field and engineering teams visualize and manage drone-captured 3D models and imagery.
- Streamlined data flow across AWS S3 and MongoDB, and built custom 3D measurement tools with Three.js for tower inspections.
- Containerized and deployed the application with Docker, AWS ECR, and ECS, standardizing the release process.

Comprehensive Platform for Mining Site Monitoring

- Architected a coal mining site monitoring platform - React.js/Redux on the frontend, Node.js/Express on the backend - to process and analyze 3D drone imagery.
- Engineered the data layer across MongoDB and Google Cloud Storage, and built 3D tools for measuring distance, area, and volume.
- Automated geospatial processing with Python and GDAL, converting .tiff to .png and rendering dynamic zoom-based tile overlays on Google Maps to visualize site changes over time.

Warehouse Monitoring System

- Co-built 3D point cloud segmentation tools (React.js, Redux Toolkit) on top of an Express.js/Python Flask backend for warehouse space analysis.

NavGurukul - Student Mentor Nov 2021 - Jul 2022

Dharamshala, Himachal Pradesh

- Trained in web development through an intensive program, then mentored 100+ students in the same curriculum as an instructor within an NGO setting.

PROJECTS

GYMAXIS - Multi-Tenant Gym Management SaaS | Flutter, NestJS, TypeScript, PostgreSQL, Sequelize, Passport.js, REST APIs

Personal SaaS product, currently in active use by a gym. Led the Flutter/NestJS rebuild of a multi-tenant gym management platform, covering role-based apps for gym owners (tenants), trainers, and members backed by a shared REST API and PostgreSQL/Sequelize data layer, with self-serve tenant signup and per-tenant branding.

- Designed and implemented secure authentication flows including native Google Sign-In (NestJS + Passport.js, ID-token verification) and a from-scratch WhatsApp OTP login/registration system built directly on the Meta WhatsApp Cloud API, including rate-limiting, hashed/expiring codes, and graceful dev-mode fallbacks.
- Built self-serve, multi-tenant onboarding flows (email/password, Google Sign-In, WhatsApp OTP) and a public "Explore Gyms" discovery experience with gym profile pages, cover-image uploads, and dynamic per-tenant theming.
- Implemented a full financial and operations module (ledgers, expenses, member subscriptions, notifications) with bug fixes driven by root-cause analysis of state serialization issues in the mobile data layer.
- Diagnosed and resolved a hard native Android OAuth bug (missing serverClientId causing silent null ID tokens) through systematic on-device debugging with adb/logcat, restoring a broken sign-in flow to full working order.
- Owned features end-to-end: Sequelize migrations and NestJS services/controllers on the backend, Riverpod state management and custom Flutter UI on the frontend, and native platform configuration (Android OAuth/Firebase) connecting them.

GrayHawk - GIS / Geospatial Analytics Platform | React 18, Flask, GDAL, Rasterio, MongoDB, Docker | greyhawkmulti-scope.com

Freelance project for a US client. Built a full-stack GIS platform for drone/aerial survey analytics, turning LAS point clouds, GeoTIFF DEMs, Shapefiles, and GeoJSON into interactive maps, 3D terrain views, elevation profiles, and volume/stockpile measurements.

- Developed geospatial processing pipelines in Python (GDAL, Rasterio, GeoPandas, Shapely, LasPy, PyProj) to compute volumes, generate contour lines, and perform cut-and-fill/stockpile analysis over user-drawn polygons.
- Implemented interactive 3D terrain and point-cloud rendering (Potree/Three.js) and multi-layer 2D mapping (Mapbox GL, Leaflet, Google Maps) with custom drawing, annotation, and measurement tools.
- Designed a RESTful Flask backend (MongoDB via MongoEngine) with JWT-based authentication, Axios token-refresh interceptors, and Bcrypt password hashing.
- Built a multispectral analytics module for comparing crop/terrain health indices across dated site visits, with data-visualization dashboards and PDF/report export.
- Containerized and deployed the multi-service stack with Docker Compose (React, Flask API, S3 proxy, MongoDB, Nginx) with automated Let's Encrypt/Certbot SSL management.

TECHNICAL SKILLS

Languages: HTML/HTML5, CSS3, JavaScript/ES6+, TypeScript, Dart, Python (NumPy, Pandas, SciPy, Matplotlib, scikit-learn)

Frameworks/Libraries: Node.js, NestJS, React.js, Express.js, Redux/Toolkit, Flutter, Riverpod, go_router, Flask, Bootstrap, jQuery, Electron.js

Geospatial/Mapping: GDAL, Rasterio, GeoPandas, Shapely, LasPy, PyProj, Mapbox GL, Leaflet, Potree, Three.js, Google Maps API

Cloud Services: AWS (S3, Lightsail, EC2, ECS, ECR), GCP (Compute Engine, Cloud Storage, Pub/Sub), Firebase

Industry Tools: Git/GitHub, VS Code, Postman, Mongoose, MongoEngine, Sequelize, Docker, Docker Compose, Nginx, JWT/Passport.js

Databases & Other: MongoDB, PostgreSQL, MySQL, Redis, RESTful APIs, Google Sign-In, Meta WhatsApp Cloud API

EDUCATION

Amity University - MCA in AI/ML (Feb 2027) | **Calcutta University** - B.Sc. (73%) (Oct 2020)